

Assignment 1

Course Code : CSE0613322

Course Title : Compiler Lab

Task : Solve ALL 3 Problems

Submitted to : SOA, MTT

Submitted by : 0432410005101088

Dept. : CSE

Batch : 55

Section : 6B1

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Code:

Problem 1: Array – Find the Missing Number

[C]

```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

int main()
{
    int n;
    printf("Input n: ");
    scanf("%d",&n);
    int arr[n];

    for(int i=0; i<n; i++)
    {
        scanf("%d",&arr[i]);
    }
    int size = sizeof(arr) / sizeof(arr[0]);
    int sum = n*(n+1)/2;

    for(int i=0; i<n; i++)
    {
        sum = sum - arr[i];
    }

    printf("%d",sum);
    return 0;
}
```

Problem 2: String – First Non-Repeating Character

[C++]

```
#include <iostream>  
#include <string>  
#include <unordered_map>  
  
using namespace std;  
  
int main()  
{  
    string text;  
    cin >> text;  
    unordered_map<char, int> freq;  
  
    for (char c : text)  
    {  
        freq[c]++;  
    }  
  
    // looping through text and matching the count in map  
  
    for (char c : text)  
    {  
        if (freq[c] == 1)  
        {  
            cout << "First non-repeating character: " << c << endl;  
            break;  
        }  
    }  
  
    return 0;  
}
```

Problem 3: Array + String – Group Anagrams

[PYTHON]

```
Anagrams = ["eat","tea","tan","ate","nat","bat"]
```

```
copy = Anagrams.copy()
```

```
grouped = []
```

```
gr_cnt = 0 # index for `grouped` list
```

```
grouped.append([])
```

```
track = [] # tracking the added values
```

```
for i in range(len(copy)): # sorting strings within list
```

```
    text = copy[i]
```

```
    sorted_text = "".join(sorted(text))
```

```
    copy[i] = sorted_text
```

```
# now `copy` is sorted but `Anagrams` is not. But they have the same  
index for the corresponding elements inside them
```

```
for i in range(len(Anagrams)):
```

```
    flag = False
```

```
    for j in range(len(Anagrams)):
```

```
        if Anagrams[j] not in track: # skips added values after first loop
```

```
            if copy[i] == copy[j]:
```

```
                grouped[gr_cnt].append(Anagrams[j])
```

```
                track.extend(grouped[gr_cnt])
```

```
            if i!=len(Anagrams)-1 and len(grouped[gr_cnt])>0: # skip when last  
loop reached
```

```
                gr_cnt=gr_cnt+1
```

```
                grouped.append([])
```

```
print(grouped)
```